

Installation Guide

Enterprise Server Deployment and Operating System Installation

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Operating System: Ubuntu Server

1. Project Overview

This project focuses on setting up an Ubuntu Server virtual machine for ABC Startup Solutions. The server will be used as a foundation for services such as file sharing, remote administration, database hosting, web hosting, and other internal services.

The installation process includes preparing the virtual machine, installing Ubuntu Server, creating the administrator account, setting the hostname, enabling SSH, checking the network connection, and updating the operating system.

2. System Requirements

Before starting the installation, the computer and virtual machine should have enough resources to run Ubuntu Server properly.

Recommended Virtual Machine Settings

Component	Recommended Setting
CPU	2 or more processor cores
Memory	4 GB RAM or higher
Storage	25 GB or more
Network	NAT or Bridged Adapter
ISO	Ubuntu Server ISO image

3. Virtual Machine Preparation

1. Open the virtualization software such as VirtualBox or VMware.
2. Create a new virtual machine and select Linux/Ubuntu as the operating system.

3. Assign the recommended RAM, processor, and storage.
4. Attach the Ubuntu Server ISO file to the virtual machine.
5. Set the network adapter to NAT or Bridged mode depending on the required setup.

4. Ubuntu Server Installation

1. Start the virtual machine and boot from the Ubuntu Server ISO.
2. Select the preferred language and keyboard layout.
3. Configure the network connection and confirm the detected interface.
4. Choose the storage option and create the required disk layout.
5. Enter the server name or hostname.
6. Create the administrator username and password.
7. Select the option to install OpenSSH Server so the server can be managed remotely.
8. Review the installation settings and continue with the installation.
9. Wait for the installation to finish, then restart the virtual machine.

5. Initial Server Configuration

After Ubuntu Server starts, log in using the account created during installation. The first step is to update the system packages.

Update the package list and installed packages:

```
sudo apt update  
sudo apt upgrade
```

Check the hostname and network information. Test the connection by using a simple ping command.

```
hostname  
ip addr  
ping -c 4 google.com
```

6. SSH Configuration and Testing

SSH allows the administrator to connect to the Ubuntu Server remotely. If OpenSSH Server was not installed during setup, it can be installed using the following command:

```
sudo apt install openssh-server
```

Check whether the SSH service is running:

```
sudo systemctl status ssh
```

7. Verification

After completing the installation and basic configuration, the server should be checked to make sure the important parts are working.

- The Ubuntu Server boots normally.
- The correct hostname is displayed.
- The server has a valid IP address.
- Internet or local network connectivity works.
- SSH service is active.
- System packages are updated.

8. Conclusion

The installation was completed by creating a virtual machine, installing Ubuntu Server, configuring the basic settings, enabling SSH, and testing the network connection. This activity helped me understand the basic process of deploying and preparing a server. It also gave me practical experience with server administration tasks that can be useful in future IT work.